

1. Research Question

How do consumers trade off price, sustainability, and store characteristics when choosing between fashion retailers, and how do individual characteristics influence store engagement relative to the opt-out?

2. Data

Respondents are MLU students, recruited via online survey. After removing incomplete responses, the dataset consists of 319 individuals who completed all 6 choice tasks (1,914 total observations).

- Store choice: Store 1, Store 2, or opt out
- Panel data structure: 6 choices per respondent
- Each person has their associated variables
- Each store choice includes store and pricing information

3. Methodology

Model Specification: Multinomial Logit (MNL) with Random Utility Maximization framework

The MNL model is appropriate for this research context because:

1. **Discrete choice experiment (DCE) structure:** Respondents choose among three alternatives (Store 1, Store 2, or opt-out)
2. **Independence of Irrelevant Alternatives (IIA) assumption:** The ratio of choice probabilities between any two alternatives is independent of other available alternatives. Tested via Hausman-McFadden test.
3. **Random Utility Theory assumption:** Respondents have Utility-maximizing behavior
4. **Panel data structure:** Likelihood contributions are computed at the individual level, multiplying choice probabilities across all 6 choices before taking the log

Post-estimation Analysis: Willingness-to-pay (WTP)

- **Formula:** $WTP = -\beta_{attribute} / \beta_{price}$
- **Interpretation:** Monetary amount an individual is willing to pay per unit increase

3. Limitations & Reflections

Sample & External Validity: The sample consists exclusively of MLU students, a group that is younger, more highly educated, and more sustainability-aware than the general population. Results may not generalise to broader consumer segments such as older shoppers, lower-income households, or those with limited online purchasing experience. Additionally, the product was held constant to a standardised black t-shirt, which limits transferability to higher-involvement or fashion-forward purchases where brand identity and aesthetic considerations play a larger role.

Hypothetical Situation Bias: Respondents may overstate their preferences for sustainability and local brands when not spending real money.

No Unobserved Heterogeneity: MNL assumes all taste variation captured by observed demographics. Mixed logit allows for random individual-specific preferences beyond observables. Assuming homogeneous attribute sensitivities across the population. A Mixed Logit model would allow coefficients to vary across individuals and provide richer insights into preference heterogeneity.

Why Not Mixed Logit?: While random coefficient models would capture unobserved preference heterogeneity, they require 10–20 choice observations per individual for stable estimation. With only 6 choices per respondent, attempting to estimate both the mean and standard deviation of each attribute coefficient across individuals would produce unreliable estimates due to weak identification. Apart from being a different model, the data is insufficient for stable estimation of the full coefficient distributions that Mixed Logit requires.

4. CONCLUSION

The MNL model is the established approach for DCE analysis, grounded in McFadden's Random Utility Theory framework. The main limitation is the assumption of homogeneous preferences, which could be relaxed with Mixed Logit in future work. Despite limitations inherent to stated preference methods, the analysis provides valuable insight into the relative importance of store-specific attributes to the consumer. Additionally, these can be expressed as monetary values through WTP analysis, offering an intuitive interpretation of the value of store attributes.